INCH-POUND

CCC-C-428G
INTERIM AMENDMENT 1
October 31, 1990

INTERIM AMENDMENT

TO

FEDERAL SPECIFICATION

CLOTH, DUCK, COTTON; FIRE, WATER, WEATHER, AND MILDEW RESISTANT

This interim amendment was developed by the U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5014 based upon currently available technical information. It is encouraged that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this interim amendment as a valid exception to CCC-C-428G, dated March 9, 1987.

PAGE 1

1.2.1, under Type II: Add the following:

"Type III - Cloth finished in Tan 380
Type IV - Cloth finished in Tan 686"

PAGE 2

2.1, under Federal Specifications: Add the following:

"CCC-C-950 - Dyeing and Aftertreating Processes for Cotton Cloths"

Under Federal Standards: Add the following:

"FED-STD-595 - Colors Used in Government Procurement"

PAGE 4

3.4.1.1: Delete and substitute the following:

"3.4.1.1 Shuttleless loom. When the cloth is woven on shuttleless looms, the fringed selvages shall be secured by a leno-lock stitch and the protruding fringe shall not exceed 3/16 inch."

AMSC N/A FSC 8305

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Add the following new paragraph and tables:

"3.4.5.3 Spectral reflectance. When olive green shades or olive drab shades are specified the spectral reflectance factors (in percent) for visible/near infrared wavelength range of 600 to 860 nanometers (nm) of the finished cloth shall meet the requirements specified in table IIIA. When Tan 380 or Tan 686 matching color chip 33446 of FED-STD-595 is specified, the spectral reflectance factors (in percent) for visible/near infrared wavelength range of 700 to 860 nm of the finished cloth shall meet the requirements specified in table IIIB. Testing shall be as specified in 4.3.3.

TABLE IIIA. Spectral reflectance values (percent) for olive green or olive drab shades

Wavelength, Nanometers (nm)	Reflectance Min.	Max.	Wavelength Nanometers (nm)	Reflectance Min.	Max.
600	3	12	740	7	52
620	3	12	760	11	60
640	3	12	780	17	64
660	3	13	800	24	70
680	3	15	820	32	80
700	3	28	840	37	86
720	5	40	860	40	88

TABLE IIIB. Spectral reflectance values (percent) for Tan 380 and Tan 686

Wavelength, Nanometers (nm)	Reflectance Min.	Max.
700	45	65
720	45	65
740	45	65
760	45	65
780	45	65
800	45	65
820	45	65
840	45	65
860	45	65"

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3.4.10 and 3.4.10.1: Delete and substitute the following:

"3.4.10 Mildew resistance. The cloth shall be mildew resistant treated in accordance with the methods specified in CCC-D-950. Unless otherwise specified, types I and II cloth shall be treated with inhibitor (d) or (e) (copper-8-quinolinolate) and types III and IV cloth shall be treated with inhibitor (a) (2,2' methylene-bis- (4-chlorophenol)) or (h) (a mixture of Zinc salts of dimethyl dithiocarbamic acid and 2 mercaptobenzothiazole)."

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Table IX, under characteristic for "Colorfastness to accelerated weathering": Add:

"Spectral reflectance ----- 3.4.5.2 ----- 4.3.3"

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Table IX: Delete "Mildew resistance": and the related tests thereunder.

Delete footnote 8/ in entirety.

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Delete footnote 9/ in entirety.

PAGE 17

Add the following paragraph:

"4.3.3 Spectral reflectance measurements in the visible/near infrared. Spectral reflectance data shall be obtained from 600 to 860 nanometers (nm) for the Olive Green and Olive Drab shades and from 700 to 860 nm for the Tan 380 and Tan 686, at 20 nm intervals on a spectrophotometer (see 6.9) relative to a barium sulfate standard, the preferred white reference standard. Other white; e.g. Halon, magnesium oxide, or vitrolite tiles (see 6.10). The spectral band width at 860 nm shall be less than 26 nm. Reflectance measurements shall be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode is used, the spectrophotometer shall operate with the specimen diffusely illuminated with full emission of a continuous source that simulates in the visible spectrum either CIE Source A or CIE Source D65. The specimen shall be measured as a single layer backed with two layers of the same fabric and shade. Readings shall be viewed at an angle no greater than 10 degrees from the normal. Photometric accuracy of the spectrophotometer shall be within 1 percent and the wavelength accuracy shall be within 2 nm. When the measured reflectance values for any color at four or more of the listed wavelengths do not meet the limits specified in tables IIIA and IIIB, it shall be considered a test failure."

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6.1.1: Delete "Types I and II." and substitute "All types."

Add the following new paragraph:

"6.1.1.3 Tan 380 and Tan 686. Unless otherwise specified, Tan 380 shall be used except for tarpaulins, truck covers and upholstery where Tan 686 shall be used.

6.2, g.: Delete "(see 3.4.10.1)" and substitute "(see 3.4.10)".

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Add the following paragraphs:

"6.9 Spectrophotometers. Suitable spectrophotometers for measuring spectral reflectance in the visible/near infrared are the Diano Hardy, Diano Match Scan, Hunter D54P-IR and Macbeth 1500 with IR option.

6.10 White standard. Barium sulfate of suitable quality for use as a white reference standard is available from the Eastman Kodak Company. The same source has available, magnesium reagent (ribbon) and Halon. Suitable tiles can be obtained from the National Insititute of Standards and Technology or the instrument manufacturers."

Preparing activity:

Army - GL

(Project 8305-A416)

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